

Laboratory 5: Hardware Add and Subtract [8 bit] with State Machine

1 INTRODUCTION

In the last lab you created a basic 3 bit add/sub circuit that used the DE1 SoC switches for inputs A and B. Due to the fact that the system did not keep track of any 'state', both A and B inputs had to be selected at the same time, thus limiting the size of the add/sub circuit. In this lab we will make use of the concept of 'state' and enter values of A and B one at a time. You will be creating an 8 bit add/sub circuit and displaying the result on the seven segment display.

1. On reset, shall enter the 'input_a' state and input 8 bits for the first input of the add/sub circuit via the slide switches and display the value on 3 seven segment displays.
2. Upon pressing a button, shall enter the 'input_b' state and input 8 bits for the first input of the add/sub circuit via the slide switches and display the value on 3 seven segment displays.
3. Upon pressing a button again, shall enter the 'disp_sum' state and the sum of A and B shall be displayed on 3 seven segment displays.
4. Upon pressing a button again, shall enter the 'disp_diff' state and the difference of A and B shall be displayed on 3 seven segment displays.
5. Upon pressing a button again, shall enter the 'input_a' state and operate as described above.
6. Shall have an active high reset switch.
7. Shall operate with unsigned base 512 numbers. Ex. $0 - 1 = 511$
8. All inputs shall be synchronized with the 50 MHz clock.
9. All 3 seven segment displays shall display a maximum decimal number of 999. [Although 512 is max realistically].
10. Shall display the present state via 4 LEDs.

2 PRE-LAB [BLOCK DIAGRAM]

You shall create a block diagram with a state transition diagram accurately mapping out your design. Also bring a printout of the block diagram to your lab class for an open discussion. Make sure to include proper synchronization, edge detection, and signal names in your diagram as well as a clear notation indicating the bit width of various busses. A state transition diagram is also required.

3 SIMULATION

Create a simulation that tests out the below use cases. You do not need to use assertions for this lab. I have posted my double_dabble.vhd file which is a process that converts a 12 bit binary number to 3 seven segment display nibbles. You will need this code to adequately display the std_logic_vectors via the 3 seven segment displays.

A	B	SUM	DIFF
5	2	7	3
2	5	7	509
200	100	300	100
100	200	300	412

4 HARDWARE

Target your design onto the DE1 SoC and receive a signoff. To save you time, I posted my compile.tcl file that lists the pin assignments that I used.

5 LAB UPLOAD => REQUIRED

In order to get full credit for the lab, please do the following.

- ☐ Delete the **work** folder that modelsim generates and the **project** and **output_files** folders in the hw folder.
- ☐ Zip up the high level lab folder
- ☐ Post the .zip file to the below link

<https://u.pcloud.com/#/puplink?code=EowXZRHG6Wbwm7eVJdGigjO2YERC12lf7>

6 SIGNOFFS – LAB 5 NAME: _____

Category	Initials	Date	Points
Block Diagram			/20
Simulation			/40
Demonstration			/40
Final Grade			/100